In the name of Almighty Allah…



Information Security

Muhammad Bilal, Ph.D.

Ph.D. Computer Science and Technology, Zhejiang University, China MS Computer Science, NTU Faisalabad Pakistan

B.Sc (Hons) Computer Science, UET Lahore Pakistan

GC University, Faisalabad

Motivation


فَبِاَيِّ اٰلَآءِ رَبِّكُمَا تُكَذِّبٰنِ
تو (اے جن و انسان!) تم دونوں اپنے رب کی کون کون سی نعمتوں کو جھٹلاؤ گے؟
SURAH REHMAN 55:13

Cryptography

Generically classified along three independent dimensions

- **Type of operations used for transforming plaintext to ciphertext**
- Number of Keys used
- Way in which plaintext is processed

Cryptography

Substitution	Transposition
❖ Caesar Cipher	❖ Rail Fence
❖ Monoalphabetic Cipher	❖ Row Column Transposition
❖ Playfair Cipher	
❖ Hill Cipher	
❖ Polyalphabetic Ciphers	
❖ One-Time Pad	

Cryptography

1. The type of operations used for transforming plaintext to ciphertext

Two general principles:

Substitution An element in plaintext is mapped into another element

Transposition Elements in the plaintext are rearranged

Fundamentals requirements are:-

- All operations are reversible(No Information lost)
- Most systems referred to as use a combination of substitution and transposition

Substitution

- A classical encryption techniques
- Where letters of plaintext are replaced by other letters or by numbers or symbols
- If plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Caesar Cipher

- Earliest known substitution cipher proposed by Julius Caesar
- First attested use in military affairs
- Replaces each letter by 3rd letter on
- C = E (3 , p) = (P + 3) mod 26
- example

m	e	e	t		m	e		a	f	t	e	r		t	h	e		p	a	r	t	Y
P	H	H	W		P	H		D	I	W	H	U		W	K	H		S	D	U	W	B

Caesar Cipher

- **We can define transformation as:**

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- Mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

then have Caesar cipher as:

$c = E(k, p) = (p + k) \bmod (26)$

$p = D(k, c) = (c - k) \bmod (26)$

Cryptanalysis of Caesar Cipher

- Only have 26 possible ciphers
  - A maps to A,B,..Z
- Easily breakable by performing **brute force search**
- Given ciphertext, just try all shifts of letters
- Recognize the plaintext
- e.g. break ciphertext "GCUA VQ DTGCM“ , where K=2(Shifting of letters)

Caesar Cipher(Examples)

1. Each letter in the plaintext message has been shifted 3 letters down in the alphabet (where K=3)

Plaintext: Honesty is the best Policy



Ciphertext: KRQHVWB LV WKH EHVW SROLFB

2. Plaintext : Knowledge is power

Ciphertext: NQRZOHGJH LV SRZHU

Caesar Cipher(Examples)

1. Each letter in the Ciphertext message has been shifted 3 letters Upward in the alphabet (where K=3)

Ciphertext: EHVW IULHQG



Plaintext: best friend

2. Ciphertext : SDNLVWDQ LV RXU FRXQWUB

Plaintext: pakistan is our country

Monoalphabetic Cipher

- Mono means one
- Rather than just shifting the alphabet
- Could shuffle (jumble) the letters arbitrarily
- Each plaintext letter maps to a different random ciphertext letter
- Hence key is 26 letters long

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Plain:  a  b c d  e f  g  h  i j k l m n o p q r s t u v w x  y z
Cipher: D  K V Q  F I  B  J  W P E S C X H T  M Y A U O L R G Z N
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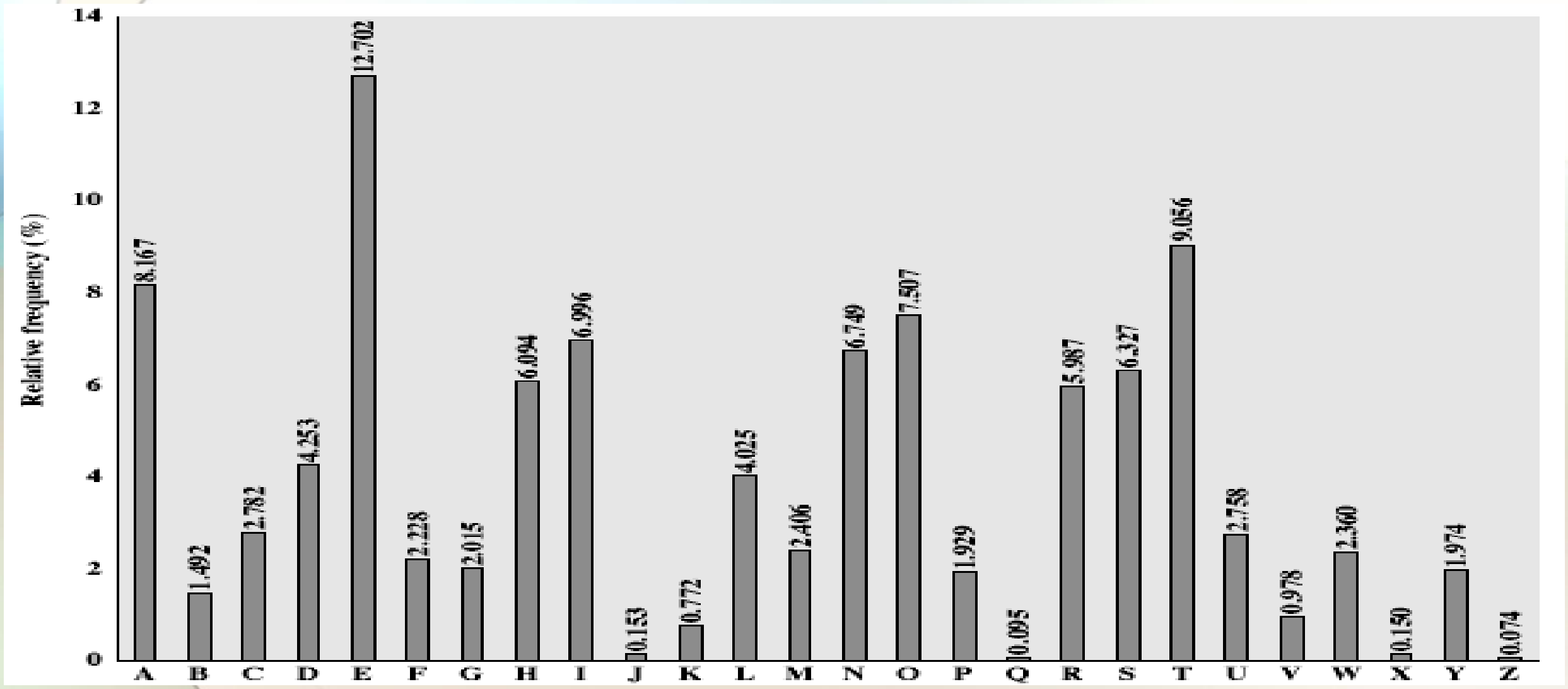
Plaintext: if we wish to replace letters

Ciphertext: WI RF RWAJ UH YFTSDVF SFUUFYA

Monoalphabetic Cipher Security

- Have a total of 26! = 4 x 1026 keys
- With so many keys, might think is secure
- But would be **!!!WRONG!!!**
- Problem is language characteristics
- Human languages are **redundant**
- letters are not equally commonly used
- In English **e** is by far the most common letter then T,R,N,I,O,A,S
- Other letters are fairly rare c,f. Z,J,K,Q,X

English Letters Frequencies



Reference

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2 Douglas Stinson ,(1995).Cryptography -Theory and Practice Boca Raton.

3 https://www.geeksforgeeks.org/computer-networks/cryptography-and-its-types/